

necessary, repaired and replaced with new ones and then reassembled to build up a product with quality that is as good as new. A remanufactured product is often cheaper than a new one as the reprocessing and manufacturing expenditures (time, energy, cost) are much less. For example, a used machine can be remanufactured for 50-60 per cent of the cost of a new machine," says Tiwari.

Green manufacturing has been successfully operating in Europe and the US. There are about 73,000 firms engaged in remanufacturing in the US. In India, unfortunately, product recovery is still not well known in the business sector. Although Indian manufacturers have realised the seriousness of the clean environment issue, no organised effort has yet been made in this field. Though there is a strong second-hand market and an unorganised sector for product recovery practices exist in India, the country has enough potential for an organised sector for remanufactured and recovered products and components.

Tyagi believes it is increasingly becoming mandatory for more and more products to meet the strict emission regulations. For instance, small engines, especially carburetted two-stroke engines, are major sources of air pollution. "To help small engine makers meet stringent new emission regulations, Freescale designers have created a chip to control all analogue functions and reduce emissions from small petrol engines," says Tyagi.

Going green is profitable

Companies that take proactive steps towards a clean and green environment through green manufacturing processes have advantages in business, says Tiwari. "They not only gain a 'greener' image in the public eye but also get more time to develop methods to reduce e-waste and can do so on their own schedule. Companies that wait until they are forced to change and switch to green operations often find themselves in an expensive, last-

How green are the firms

While companies claim to provide customers with green gadgets, how can we be sure exactly where our favourite companies stand in their overall switch to green practices? Let's find out.

Motorola's manufacturing facility at Sriperumbudur near Chennai was honoured with the Leadership in Energy and Environmental Design (LEEDR) certification by the Indian Green Buildings Council. The award recognises Motorola's factory as the first green manufacturing site in India, which manages storm water run-off and has a rainwater harvesting pond. The facility also has a water-recycling facility by which it is able to reduce municipal water consumption.

Samsung Electronics announced a green management initiative that, the company claims, would establish it as a leading eco-friendly company by 2013. Focused on achieving low-carbon economic growth, the company laid out four core green management objectives as part of its 'Eco-Management 2013' plan. The objectives included reducing greenhouse gas emissions by cutting total indirect greenhouse gas emissions from all products by 84 million tons over a five year period; ensuring all Samsung products are eco-friendly enough to meet global eco-mark standards; investing 5.4 trillion Korean Won in various eco-management initiatives; and enhancing green partnerships with suppliers and partners.

Wipro Infotech has been rated India's number one green brand and among the top five global green brands by Greenpeace. Wipro claims to have made sustainable efforts on energy-efficiency and effective e-waste management programmes. It not only uses eco-friendly technologies but practices all green manufacturing processes.

Epson makes its packaging from trees grown specifically for that purpose and plants 20 per cent more trees than it cuts down.

Lenovo's energy-efficient monitors contain 25 per cent recycled material, and the company recovers gold, silver and other precious metals from products at the end of their lives.

Nokia has reduced the amount of no-load energy (the electricity that cellphone chargers consume when they are left plugged in after a phone is charged) sucked up by its phone chargers by 90 per cent, and now its phones alert users to unplug chargers.

Panasonic's plasma TVs consume 96 per cent less energy than its plasma TVs did in 2000.

Kodak has recycled 1.2 billion single-use cameras since 1990, recycling 120 million last year. Almost all the new single-use cameras of the company contain recycled parts.

Intel runs a capital funding programme for conservation and efficiency efforts, approving more than 200 projects since 2001 that have saved more than 400 million kW hours of electricity.

minute scramble to meet the requirements. As a result, they end up investing more money for a less effective solution," he says.

Green manufacturing has become not just a more responsible way of producing goods but a more profitable one as well. Green production is a business strategy that focuses on profitability through environment-friendly operating processes. Proponents of this management philosophy believe that green production is a sensible course to follow not only because of the benefits it bestows on the environment but also

because of its fundamental strategic soundness.

"The only problem that hinders green products from becoming popular is the fact that they are more expensive than the traditional products, so it is difficult to create a market for them," says BenQ spokesperson. "But ultimately, manufacturers who have invested money in greener production processes have been able to cut the cost of their green products," concludes Tyagi. ●

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